

Chapter 8 Assignment — Polar Coordinates and Parametric Equations

Name: _____

Date: _____

How to use this set: Each topic begins with *Strategy Notes* and a *Worked Example*. Then try the *Practice* (skill) and the *Challenge* (deeper) items. Give exact answers using radicals and π when possible. For polar angles, state units and choose $0 \leq \theta < 2\pi$ unless otherwise specified.

1. Topic 1: Polar Coordinates and Conversions

Strategy Notes. Use $x = r \cos \theta$, $y = r \sin \theta$, $r = \sqrt{x^2 + y^2}$, and $\tan \theta = \frac{y}{x}$ with quadrant checks. Remember that $(r, \theta) = (-r, \theta + \pi)$ gives the same point.

Worked Example. Convert the rectangular point $(-5, 0)$ to polar form.

Solution. $r = \sqrt{(-5)^2 + 0^2} = 5$. Since the point is on the negative x -axis, $\theta = \pi$. One representation is $(5, \pi)$. An alternative is $(-5, 0)$.

Practice.

- (a) Convert $(6\sqrt{2}, 6)$ to polar coordinates with $r > 0$ and $0 \leq \theta < 2\pi$.

- (b) Convert the polar point $(9, -180^\circ)$ to rectangular coordinates.

Challenge.

- (a) The rectangular point $(-6, 2\sqrt{3})$ has many polar representations. Find one with $r > 0$ and one with $r < 0$, both with $0 \leq \theta < 2\pi$.

2. Topic 2: Graphs of Polar Equations

Strategy Notes. Recognize families: $r = a \cos \theta$ or $r = a \sin \theta$ are circles; $r = a \pm b \cos \theta$ or $a \pm b \sin \theta$ are limaçons (inner loop if $|a| < |b|$); $r = \sin(k\theta)$ or $r = \cos(k\theta)$ are rose curves (if k is even, $2k$ petals; if k is odd, k petals). Converting to rectangular can reveal the type quickly.

Worked Example. Graph $r = 8 \cos \theta$ and classify the curve.

Solution. Multiply by r : $r^2 = 8r \cos \theta \Rightarrow x^2 + y^2 = 8x$, so $(x - 4)^2 + y^2 = 4^2$. This is a circle of radius 4 centered at $(4, 0)$.

Practice.

- (a) Graph $r = 3 + 6 \sin \theta$. Identify the curve type and indicate intercepts with the coordinate axes.

- (b) Graph $r = \frac{4}{1 - \cos \theta}$. Convert to rectangular form, identify the conic, and mark key features (vertex/focus/directrix or asymptotes as appropriate).

Challenge.

- (a) What does the graph of $r = 2 + 3\theta$ look like for $\theta \geq 0$? Sketch the first two turns and indicate the direction as θ increases.

3. **Topic 3: Complex Numbers in Polar Form and De Moivre's Theorem**

Strategy Notes. For $z = a + bi$, write $z = r(\cos \theta + i \sin \theta)$ where $r = |z| = \sqrt{a^2 + b^2}$ and $\theta = \text{Arg}(z)$. Then

$$\begin{aligned} z_1 z_2 &= r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)), \\ \frac{z_1}{z_2} &= \frac{r_1}{r_2} (\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)), \\ z^n &= r^n (\cos(n\theta) + i \sin(n\theta)). \end{aligned}$$

Worked Example. Let $z = 1 + \sqrt{3}i$. Write z in polar form and compute z^9 .

Solution. $r = \sqrt{1 + 3} = 2$, $\theta = \arctan(\sqrt{3}) = \frac{\pi}{3}$. Thus $z = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$. Then $z^9 = 2^9(\cos 3\pi + i \sin 3\pi) = 512(\cos \pi + i \sin \pi) = -512$.

Practice.

- (a) For $z_1 = 4(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12})$ and $z_2 = 2(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12})$, compute $z_1 z_2$ and $\frac{z_1}{z_2}$ in polar form.

- (b) Express $-1 - i$ in polar form (principal argument in $(-\pi, \pi]$).

Challenge.

- (a) Compute all fourth roots of $a = 1 + \sqrt{3}$ and describe where they lie in the complex plane.

4. **Topic 4: Parametric Equations and Eliminating the Parameter**

Strategy Notes. For lines, use $(x, y) = (x_0, y_0) + t(v_x, v_y)$. For ellipses $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, use $x = h + a \cos t$, $y = k + b \sin t$. Eliminate t by using identities like $\sin^2 t + \cos^2 t = 1$.

Worked Example. The curve $x = 3 \sin t + 3$, $y = 2 \cos t$ has what rectangular equation?

Solution. $\frac{(x-3)^2}{3^2} + \frac{y^2}{2^2} = \sin^2 t + \cos^2 t = 1$, an ellipse centered at $(3, 0)$ with semi-axes 3 and 2.

Practice.

(a) Find parametric equations for the line of slope 2 through $(3, 5)$.

(b) Find parametric equations for $4x^2 + 9y^2 = 144$.

Challenge.

- (a) Eliminate the parameter for $x = 3\sin(2t)$, $y = 3\cos(2t)$ and identify the curve; indicate the direction of motion as t increases.

Strategy Notes. Uniform circular motion of period T about center (h, k) and radius R can be written as $x = h + R \cos(\omega t + \phi)$, $y = k + R \sin(\omega t + \phi)$ with angular speed $\omega = \frac{2\pi}{T}$; choose phase ϕ to match the starting point and sign of rotation.

Practice.

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